

KESHAV DAYAL

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SUMMARY

Software Developer with **4+ years of experience** designing and shipping scalable web applications, distributed systems, and AI-powered tools. Comfortable across the stack from crafting responsive React and Next.js frontends to architecting Python backends with Flask and Django, containerizing with Docker, and working with PostgreSQL at scale. Has worked across federated cloud infrastructure, open-source research platforms, and developer tooling, with a track record of contributing to international research ecosystems and production-grade products.

WORK EXPERIENCE

Freelance Software Engineer (SWE-1) [@Elixir Cloud & AAI](#) | Switzerland (Remote) | Feb 2025 - Current

- Developed a full-stack TES Dashboard (Flask API + React frontend) for real-time monitoring of workflow submissions, task execution, and compute resource management across 6+ distributed cloud nodes, reducing manual status-checking overhead by **~40%** for research teams
- Established live service health monitoring, performance metrics, and batch processing interfaces across 3+ cloud providers, cutting mean time to detect infrastructure issues by **~50%**
- Enforced role-based access control with 4 permission tiers and secure authentication flows, **enabling 10+ research institutions** to manage workflow execution across multi-tenant compute pipelines
- Contributed to the GA4GH-aligned ELIXIR cloud ecosystem, supporting genomics-scale data analysis **across 8+ international research nodes**

Software Developer (SDE-1) [@Pacific Analytics](#) | Australia (Remote) | Mar 2025 - Feb 2026

- Architected and optimized a collaborative research platform spanning 4 core modules (Knowledge Base, Notes, Cohorts, Datasets), deploying lazy-loaded Jupyter environments that reduced initial load time by **~35%** and improved frontend-backend modularity
- Designed a scalable datasets API with pagination, CRUD, and error handling supporting **5+ dataset types**, alongside backend-driven profile management with ORCID integration for consistent researcher identity across institutions
- Resolved **20+ complex state** and cross-component bugs across **10+ components**, standardizing UI behavior and workflow stability for predictable platform operations
- Enhanced research output portability through optimized Markdown/PDF exports and standardized dark-mode theming, improving cross-institutional usability across **3+ deployment environments**

AI/ML Developer at [@CryptoXpress](#) | [CERTIFICATE](#) | Nov 2024 – Feb 2025

- Built AI-powered social media bots across **4 platforms (Reddit, Twitter/X, Telegram, Instagram)** using a Flask backend integrated with the Anthropic API, automating 500+ daily community interactions and reducing **manual engagement effort by ~70%**
- Engineered a persona-driven engine with **10+ JSON-based profiles** simulating human-like behavior, with deduplication pipelines tracking **1,000+ post** and comment IDs to ensure non-duplicate engagement
- Deployed Telegram bots via Telethon monitoring **15+ groups in real time**, with inactivity thresholds and controlled promotional messaging cadence for scalable outreach

Full Stack Developer at [@BurdenOff](#) | [CERTIFICATE](#) | Nov 2024 – May 2025

- Redesigned and enhanced the **91Springboard web application** using Next.js, React, and Tailwind CSS, improving Lighthouse **performance score by ~30%** and eliminating layout inconsistencies across **15+ pages**
- Built 20+ reusable UI components using modern component-based architecture, reducing average feature **development time by ~25%**
- Wired **5+ backend APIs** for smooth data flow and containerized the application with Docker, streamlining deployment workflows across development and production environments

PROJECT

AI Interview <i>NextJs, Firebase, Tailwind CSS, Vapi AI, Zod, Gemini</i>	Built a full-stack mock interview platform where an AI voice agent conducts interviews based on role, level, and tech stack. Integrated Google Gemini for automated feedback with detailed scoring across key competencies. Features Firebase authentication, personal dashboard, and responsive UI using Next.js and Tailwind CSS.	Github Website
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EDUCATION

Maharaja Surajmal Institute (GGSIPU) Bachelor of Computer Applications CGPA: 8.3	New Delhi, India Oct 2022 – May 2025
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TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL, Bash
Frontend: React.js, Next.js, Tailwind CSS, Responsive UI, Framer Motion | **Backend:** Node.js, Express.js, Django, Flask, REST APIs, API Design, MVC Architecture
Databases: MongoDB, PostgreSQL, MySQL, SQL/NoSQL Schema Design, Indexing | **Cloud & DevOps:** Docker, Kubernetes, Git, GitHub, Linux, CI/CD, Containerization
Systems & Data: Microservices, Event-Driven Systems, Nextflow, Data Pipelines, Jupyter, Google Colab | **Auth & Security:** JWT, OAuth2, RBAC, Keycloak, Secure API Practices
Tools: Sanity CMS, GROQ, Postman, Figma, Canva, Blender | **Soft Skills:** Leadership, Communication, Teamwork

ACHIEVEMENT

HACKATHON ORGANIZER	- Organized a hackathon at Microsoft's Gurgaon office with 148 teams and 4000+ participants. As the Chief Advisor, collaborated with a team of 50+ students, providing guidance and support to ensure the event's success. - Organised a hackathon at Galgotias University with 100+ teams and 2000+ participants. As the Chief Organiser, Collaborated with a team of 25 Students.
HACKATHON WINNER	Achieved 3rd Rank in the Youth4Future Hackathon out of 62 teams , earning ₹5,000. Led the team in designing and implementing an innovative solution , contributing to problem solving , technical execution , and effective collaboration , demonstrating strong coding, teamwork, and project delivery skills under competitive conditions.